Remanent Magnetization, Cooling History, and Paleomagnetic Record of the Marys Peak Sill, Oregon

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A paleomagnetic investigation of the middle Oligocene (29.6 m.y.) Marys Peak sill (44°30'N, 123°35'W) demonstrates a relationship between the intrusion's cooling history and its remanent magnetization. The 390-meter-thick sill is vertically differentiated, varying in compositon from a gabbro to a diorite with associated pegmatite and aplite facies. From 7 to 10 cores per sampling site were taken at 26 sites separated vertically by 7 to 10 meters along a traverse through the sill. Directions and intensities of remanent magnetization were measured for all specimens at the original magnetization level and after alternating-field demagnetization at 100-, 200-, and 400-oe peak demagnetizing fields. Demagnetization behavior. Curie temperatures, and polished section examinations indicate that the remanence is stable. The directions of natural remanent magnetization (NRM) at individual sites are well grouped and significantly different from the present field. The site mean direction pattern changes systematically from $D \cong 20°$, $I \cong 50°$, in the lower portion to $D \cong 10°$, $I \cong 50°$, in the central region to $D \cong 25°$, $I \cong 50°$, at the top. A mathematical approximation of the cooling history permits the transformation from the geometric record of the sill to a time record of the earth's magnetic field that existed when the sill cooled. Several models were evaluated and the results indicate that the range of the paleomagnetic record is approximately 2000 to 6000 years. It is concluded that the change in remanent magnetization direction could have been caused by a large nondipole field component or an excursion of the main dipole field.

INTRODUCTION

The earth's magnetic field is subject to variation over an extremely broad time spectrum, and its analysis is based largely on direct observatory measurements [*Hide and Roberts*, 1961; *Yukatake*, 1962; *McDonald and Watson*, 1968]. These measurements date only to the sixteenth century [*Bauer*, 1895] and consequently provide no information about components of secular variation with longer periods. Paleomagnetic studies, although contributing greatly to the knowledge of earlier changes in the earth's field, suffer from the limitation that they yield ancient field directions known as a function of stratigraphy, but not generally as a function of time. A means of obtaining a continuous time record of the magnetic field, first suggested by *Jaeger and Green* [1956], is the study of the paleomagnetic record of an intrusive body. The results of such a paleomagnetic investigation of the Marys Peak sill, Oregon, are presented here. In the case of a thick intrusion, the thermoremanent magnetization (TRM) is not acquired at once but over a long period of time as the Curie temperature isotherms migrate from the outside margins inward. The result is a record of paleomagnetic direction as a function of position in the intrusion. To transform this record to one of paleomagnetic direction as a function of time, it is necessary to know the Curie temperatures through the sill and also the mutual relations of temperature, time, and position.

This concept was applied by *Jaeger and Green* [1956] to remanent magnetization measurements on a vertical bore core taken from a portion of a Mesozoic sill. They considered the sill an infinite slab that cooled symmetrically about its center from a uniform temperature. The cooling history of the Marys Peak sill is more complex because of the role played by differentiation during solidification. The temperature distribution during cooling was not symmetrical about the geometrical center of the sill, and this temperature asymmetry

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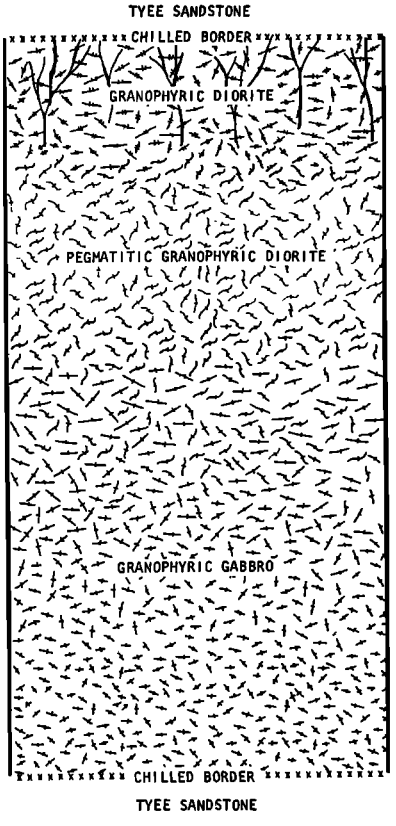


Fig. 1. Generalized geologic section of the Marys Peak sill [after *Roberts,* 1953].

is reflected in an asymmetry in the observed paleomagnetic directions. The paleomagnetic measurements will be interpreted in terms of a cooling history model developed for this asymmetry.

The Marys Peak sill (44°30′N, 123°35′W) is one of a suite of closely related mid-Tertiary gabbroic sills in the Oregon Coast Range [*Snavely and Wagner,* 1961]. A K-Ar age determination on hornblende from the pegmatite unit of the sill gives an age of 29.7 ± 1.2 m.y. (Parke Snavely, Jr., personal communication, U. S. Geological Survey). This date is in the mid-Oligocene [*Holmes,* 1960]. The Marys Peak sill intruded the Tyee sandstone conformably and horizontally beneath approximately 3000 meters of cover, as suggested by its relative degree of fractional crystallization and its stratigraphic position in the section, (Parke Snavely, Jr., personal communication). The top is essentially parallel to the base but is marked by local faults and swells. The sill is cut in several places by faults with nearly vertical displacement [*Baldwin,* 1956]. The present study was made on one fault block. The Marys Peak sill is approximately 390 meters thick and is thought to be an erosional remnant of a sill of much greater areal extent, which now caps several peaks in the vicinity. For the purpose of the mathematical model the intrusion can be approximated by an infinite horizontal slab 390 meters thick.

Parameters required to define further the cooling history model may be obtained from the following petrological data. The sill is vertically differentiated from a tholeiitic magma [*Snavely and Wagner,* 1961]. The sequence of rock types varies gradationally (Figure 1) from a coarse-grained granophyric gabbro at the base to a granophyric diorite with associated pegmatite and aplite facies in the upper portion. A micropegmatite intergrowth of quartz and orthoclase is found at all levels [*Roberts,* 1953; *Snavely and Wagner,* 1961]. It should be emphasized that the two major rock types grade into one another, so that the boundary between the phases shown in Figure 1 is somewhat arbitrary. The chilled borders of the sill are basalt. The effect of baking on the country rock is evidenced by a zone of hornfels extending a few feet above and below the chilled borders. More complete petrologic discussions are given by *Roberts* [1953] and *Snavely and Wagner* [1961].

The temperature of intrusion is a matter of conjecture. Polished sections of samples from the chilled borders show titanomagnetite grains that must have formed prior to emplacement. Since magnetite will not form above approximately 1500°C, the temperature of intrusion must have been below this value. Melting experiments by *Tilley et al.* [1963] indicate a liquidus temperature for the Skaergaard gabbroic magma of approximately 1230°C, and *Winkler* [1965] suggests 1200°C as the intrusion temperature of the typical gabbroic magma.

Differentiation by crystal settling and fractional crystallization followed the liquid intrusion [*Roberts,* 1953]. The exceptional thickness of the intrusion and great depth of burial maintained the magma in a liquid state for a

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long period of time, allowing olivine and pyroxene to become concentrated in the lower sill by crystal settling. A major portion of the granophyric gabbro formed in this way and cooled significantly before the remainder of the sill solidified. This interpretation is supported by the magnetic data. As the heavier crystals settled, the less dense liquid magma was displaced progressively upward by individual particle movement. In the later stages of solidification, the process of fractional crystallization dominated and the lighter, silica-enriched residuum became increasingly concentrated in the upper sill. This was accomplished by an upward concentration of heat through transfer of material and volatile constituents and the upward movement of the surface of crystallization [*Jaeger,* 1958]. At the time crystallization ceased, the granophyric diorite zone was very likely at a nearly uniform high temperature.

The final pegmatitic phase had an involved crystallization history. Experiments by *Tilley et al.* [1963] show that crystallization of pegmatite constituent minerals takes place over a range of 200° to 300°C. The intrasite coherence of the natural remanent magnetization (NRM) directions indicates that the remanent magnetization of the pegmatite phase was acquired after solidification. Curie temperatures measured in this region are approximately 550°C, and the ferromagnetic minerals are interstitial to the other mineral grains in the pegmatite. This indicates that solidification of this facies must have been complete above the Curie temperature. The maximum development of the pegmatite is approximately 300 meters above the base of the sill, according to field observations. These observations are confirmed by chemical analyses, which indicate a maximum silica content at this level (Parke Snavely, Jr., personal communication). It is assumed that this point was the last in the sill to crystallize and that it represents a temperature maximum at the time of final consolidation. When solidification was complete, the temperature was distributed asymmetrically about the pegmatite zone. Subsequently, the temperature distribution became more nearly symmetrical, so that the last point to acquire a magnetization was below the last point to crystallize and near the center of the sill.

Magnetic Measurements

From 7 to 10 cores per sampling site were taken at 26 sites separated vertically by 7 to 30 meters along a primary traverse and at supplementary sites at other points in the sill. Each site covers an area approximately 2.5 meters high and 15 meters wide. All cores were obtained by means of a portable gasoline-powered, water-cooled diamond core drill and were oriented in situ with an estimated accuracy of ±2° (s.d.), using a procedure described by *Doell and Cox* [1963]. The cores were 2.49 cm in diameter and were cut into specimens 2.28 cm long for remanent magnetization measurements.

Natural Remanent Magnetization

Measurements of the direction and intensity of remanent magnetization were made on a spinner magnetometer constructed in the Stanford Geophysical Laboratory after a design described by *Doell and Cox* [1965]. The magnetometer measurements were reduced and plotted by a digital computer. The direction measurements are estimated to have an accuracy of ±1.3°(s.d.), and the intensity measurements, an accuracy of ±6%(s.d.).

All specimens from a given site were demagnetized in alternating fields by using equipment described by *Doell and Cox* [1967]. The demagnetization process involves the rotation of a specimen about three perpendicular axes in an alternating field that is smoothly reduced from a peak value. Specimens showing a direction change of greater than 10 degrees following a demagnetization were inverted in the sample holder and the demagnetization repeated. The results of the two trials were averaged and given unit weight in the analysis. This procedure served to reduce any component of magnetization added by the demagnetization instrument [*Doell and Cox,* 1967].

For analysis, the natural remanent magnetization (NRM) measurements are grouped by site, and each site is assumed to represent a single point in the magnetization history of the sill (Table 1). Direction and intensity values were measured for one specimen per core for each core taken at a site. Each specimen was demagnetized in alternating fields with progressively higher peak values of 100, 200, and 400 oe

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TABLE 1. Site Average NRM Measurements for Marys Peak Sill

Position, meters	Original Magnetization					100-oe Demagnetization					200-oe Demagnetization					400-oe Demagnetization				
	N	IR	DR	k	α_{95}, deg.	N	IR	DR	k	α_{95}, deg.	N	IR	DR	k	α_{95}, deg.	N	IR	DR	k	α_{95}, deg.
388.6	7	46.6	35.9	53.1	8.4	8	53.4	21.0	22.2	12.0	*					*				
387.7	7	32.7	64.0	6.6	25.3	*					*					*				
373.4	8	40.2	55.1	9.5	19.0	8	44.1	48.8	42.0	8.6	6	44.8	39.3	155.7	5.4	7	46.6	46.7	28.3	11.5
326.1	8	36.8	10.2	27.9	10.7	8	38.4	13.3	34.4	9.6	8	42.2	14.8	42.8	8.6	8	42.3	19.5	68.0	6.8
324.6	9	−8.6	17.0	2.1	47.5	9	12.6	23.7	2.0	49.4	7	54.3	29.6	4.1	34.2	9	81.6	139.7	4.6	27.1
303.3	8	26.9	19.4	13.4	15.7	8	27.7	22.6	76.8	6.4	8	26.3	20.9	68.5	6.7	8	17.7	8.8	4.0	31.7
286.2	7	20.4	2.8	16.3	15.4	7	38.7	22.5	27.3	11.8	7	37.3	19.1	23.0	12.9	6	51.3	49.8	1.8	74.6
269.1	9	67.3	33.4	74.1	5.4	8	60.1	22.9	65.5	6.9	9	61.3	24.0	49.5	7.4	9	64.2	34.5	10.5	16.7
253.6	11	28.5	28.3	29.7	8.5	10	12.4	21.2	67.3	5.9	10	7.2	20.4	41.5	7.6	11	17.6	20.8	10.2	15.0
234.7	8	33.1	41.3	28.8	10.5	8	44.4	17.2	66.9	6.8	7	42.1	16.3	149.6	5.0	8	43.0	10.1	19.0	13.0
222.2	8	24.9	17.1	155.9	4.5	8	16.4	16.5	388.1	2.8	8	10.4	16.4	317.8	3.1	8	5.0	17.9	51.0	7.8
211.5	8	32.1	24.5	55.3	7.5	8	25.6	22.3	73.1	6.5	8	19.0	21.6	89.1	5.7	8	13.0	14.8	10.5	18.0
201.0	8	19.1	17.6	81.2	6.2	8	14.4	14.2	217.9	3.8	8	8.7	13.0	388.8	2.8	8	3.2	19.3	60.7	7.2
186.0	7	10.4	10.3	13.7	16.9	7	2.4	5.4	15.3	15.9	7	−3.1	3.7	15.8	15.6	7	−10.0	5.4	14.4	16.5
176.5	10	16.7	9.7	110.8	4.6	10	10.8	9.8	98.4	4.9	9	7.8	10.1	80.7	5.8	9	3.6	15.3	31.3	9.3
166.1	8	43.3	11.4	64.0	7.0	8	41.8	18.9	215.3	3.8	8	39.9	19.4	318.9	3.1	8	38.0	20.8	58.4	7.3
143.3	9	48.1	21.2	171.0	3.9	8	47.7	20.1	272.5	3.4	7	45.2	17.5	123.1	5.5	9	38.1	20.7	15.4	13.6
137.2	9	51.5	27.6	189.0	3.8	9	49.2	25.8	276.4	3.2	9	47.3	26.6	168.9	4.0	9	42.3	36.5	7.2	20.6
114.3	9	58.9	19.5	110.8	4.9	8	56.7	17.0	150.9	4.5	8	54.6	16.1	275.9	3.3	8	48.0	23.1	20.6	12.5
67.1	8	54.7	19.8	75.1	6.4	7	53.3	20.7	79.4	6.8	7	50.0	20.1	128.9	5.3	7	37.5	16.0	21.8	13.2
50.3	8	55.7	21.9	85.6	6.0	*					8	48.7	17.2	147.6	4.6	8	42.5	15.5	56.0	7.5
39.6	9	57.1	22.4	98.1	5.2	9	52.8	16.2	242.6	3.3	9	50.5	13.4	173.7	3.9	9	52.8	20.3	5.4	24.5
25.6	8	54.6	18.0	113.6	5.2	8	54.0	17.7	156.6	4.4	8	51.5	18.7	77.7	6.3	9	47.5	20.8	22.2	11.2
14.6	9	54.7	19.8	56.8	6.9	9	50.5	22.2	92.4	5.4	9	49.4	27.3	54.4	7.0	9	52.1	21.8	20.6	11.6
7.3	9	50.5	24.5	171.3	3.9	*					9	51.8	29.0	138.3	4.4	9	60.5	28.8	24.2	10.7
0	7	78.5	0.6	3.0	42.5	7	78.6	336.9	2.7	45.8	7	81.0	328.9	2.8	45.0	7	79.0	352.8	2.5	47.9

Notes

Position, meters above base of sill.
N, number of sample cores.
IR, site mean inclination of NRM.
DR, site mean declination of NRM.
k, intrasite precision parameter, $(N - 1)/(N - R)$ [*Fisher*, 1953].
α_{95}, 95% confidence cone about average direction of magnetization.
* Not demagnetized at this level.

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(except as otherwise noted in the table). Results for representative sites are illustrated in Figure 2. In general, the site directions are well grouped and significantly displaced from the present field direction. The fact that the scatter of directions at each site is not related to sample location suggests that the site may be considered to have been magnetized at once for the purpose of this study. The grouping of directions at the site level and the assumption that each site acquired its magnetization at essentially one point in time indicate that the Fisher statistics have meaning at this level (Table 1).

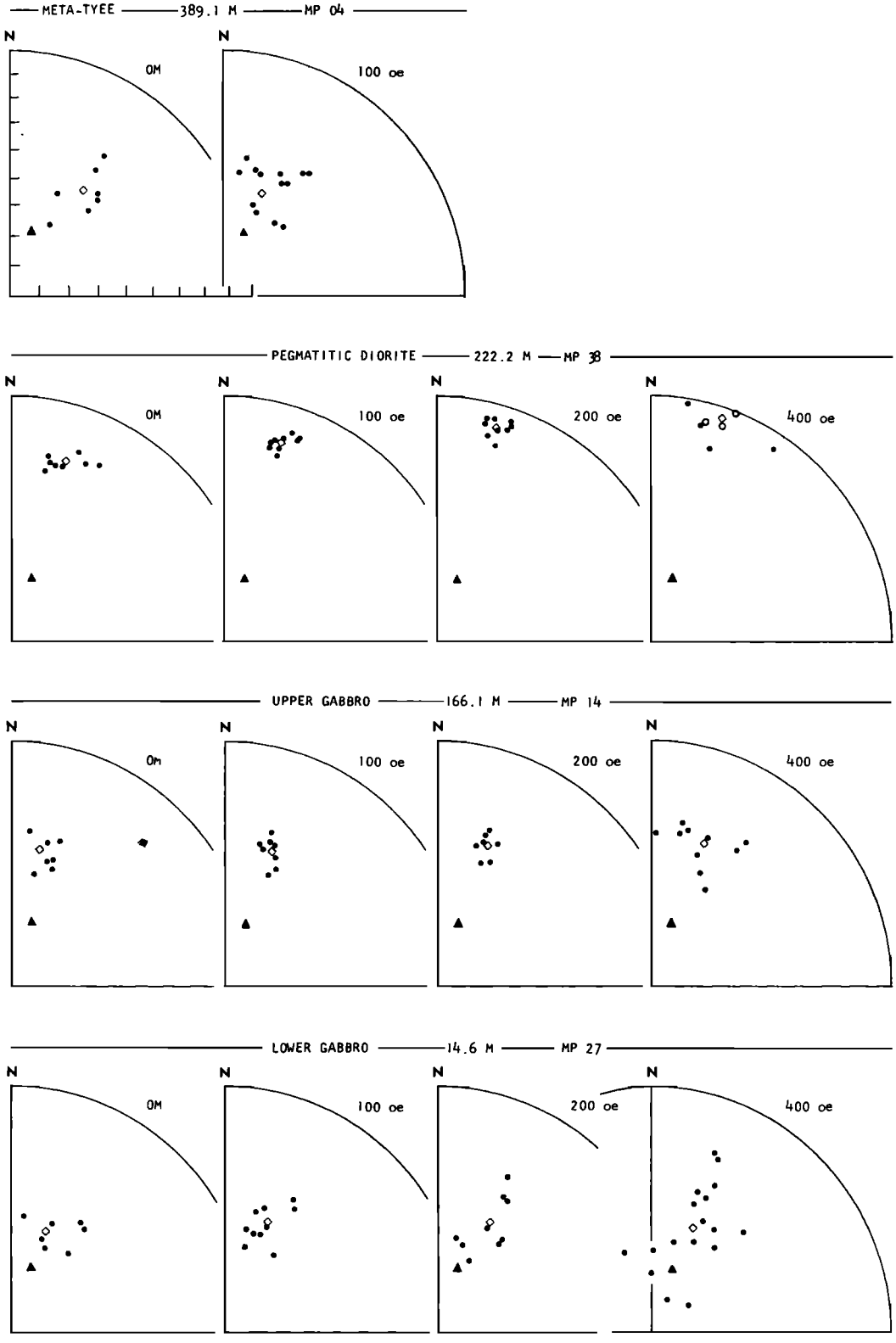
The NRM site mean direction changes systematically as a function of position in the sill, which is illustrated by a plot of remanent magnetic inclination as a function of position (Figure 3). Note that the change in NRM direction transcends rock type. Within-site scatter of remanent magnetic directions is effectively reduced by alternating-field demagnetization, and the pattern of direction change through the sill is further defined (Figures 2 and 3). The radius of the α_{95} confidence circle is an index of direction scatter, and Table 1 shows that almost all site α_{95} values decrease on demagnetization in peak fields of 100 to 200 oe. At the 400-oe level, within-site scatter of directions increases and the information is obscured because the specimen magnetization is weak relative to the anhysteretic remanent magnetization added by the demagnetization apparatus [*Doell and Cox*, 1967]. The similarity in sequence of NRM directions measured in the upper and lower sill is also more pronounced following demagnetization (Figure 4). In summary, the lower sill site mean directions are grouped and significantly displaced from the present field direction. A systematic change to a shallow inclination begins in the vicinity of 114.3 meters and continues to a minimum inclination at approximately 186.0 meters at the approximate center of the sill. The site mean direction gradually steepens again higher in the sill, returning at the top margin to the direction of the lower sill.

The sites of the upper sill show more erratic directions of magnetization than those of the gabbro phase. Fluctuations in a number of physical properties are expected in the pegmatite zone of a differentiated gabbroic sill [*Walker*, 1958] and remanent magnetization appears to be no exception. Some of the variation in direction found in this part of the sill may be, at least partly, explained by differences in Curie temperature and thus differences in the time of magnetization. The steep inclination of direction of magnetization observed at 234.7 meters is due to the exceptional inclination of two specimen values; however, the site is consistent in all other respects and is retained in the analysis. The site at 269.0 meters exhibits an anomalously steep inclination, and the chemical data for the site diverge greatly from the trend in this part of the sill (Parke Snavely, Jr., personal communication). The site is not considered in the analysis. The α_{95} value for the site at 324.6 meters is extremely large and the site is not considered in the analysis. The NRM direction data from 387.7 meters, a site that traverses the upper chilled border of the sill, are not well grouped and are not considered in the analysis. However, three cores taken from the chilled basalt margin portion of this site fall within the group of the metasediment site (388.6 meters) and, therefore, support the validity of the metasediment direction.

Additional demagnetization analysis was made to determine the composition of the NRM. Two specimens from each site were subjected to partial demagnetization in alternating fields of 12.5, 25, 50, 100, 200, and 400 oe and sometimes 600 and 800 oe. The demagnetization behavior, shown by the plots of Figure 5, does not change markedly through the sill. It is typical, although not diagnostic, of a stable TRM [*Doell and Cox*, 1967]. Individual specimen demagnetization experiments showed no systematic direction change with demagnetization. When the statistics for an entire site are considered, it is apparent that at increased demagnetization fields a small component parallel to the present earth's field is removed (Table 1 and Figure 2). This behavior is expected of a viscous remanent magnetization with a relatively long relaxation time [*Rimbert*, 1959].

Intensity of remanent magnetization changes as a function of rock type rather than NRM direction (Figure 6). The site mean intensity of NRM varies from 5×10^{-4} emu/cc in the lower portion of the gabbro to a maximum 4×10^{-3} emu/cc in the pegmatite phase. Demagnetization to 400 oe removes a larger fraction of magnetization in the pegmatite than in the gabbro,

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Fig. 2. Changes in direction of NRM with position in sill and with alternating-field demagnetization. Sites are designated by distance from sill base. North-seeking direction down, closed circle; north-seeking direction up, open circle; site mean direction, open diamond; present field direction, closed triangle. Lambert equal-area projection.

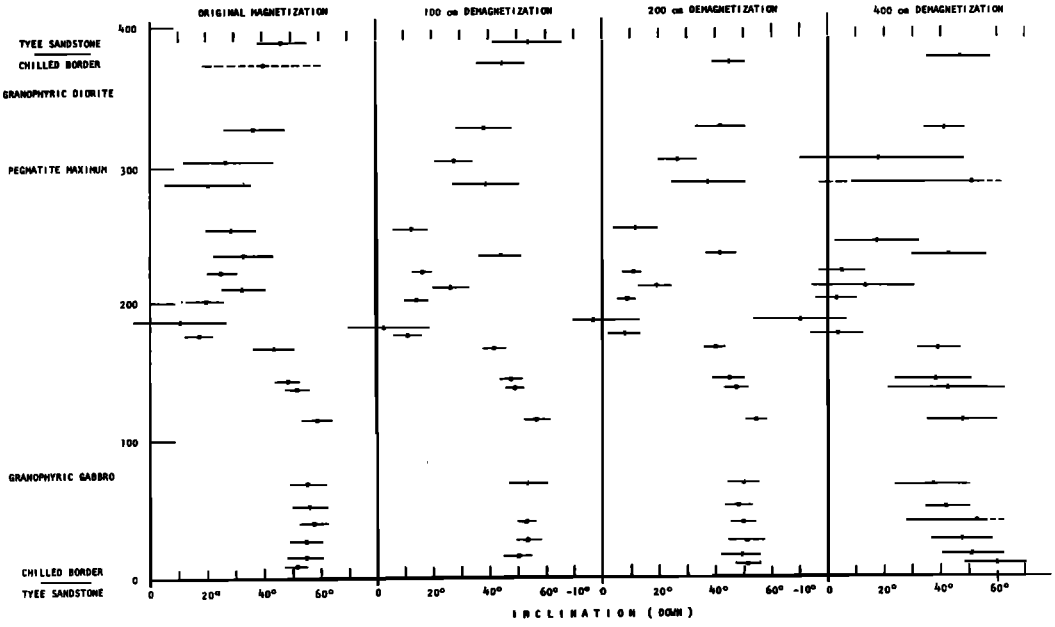


Fig. 3. Inclination of NRM direction versus position in the Marys Peak sill. Ordinate is the distance above the sill base in meters. Values given are site mean inclination with 95% confidence limits for original magnetization and after demagnetization in peak fields of 100, 200, and 400 oe.

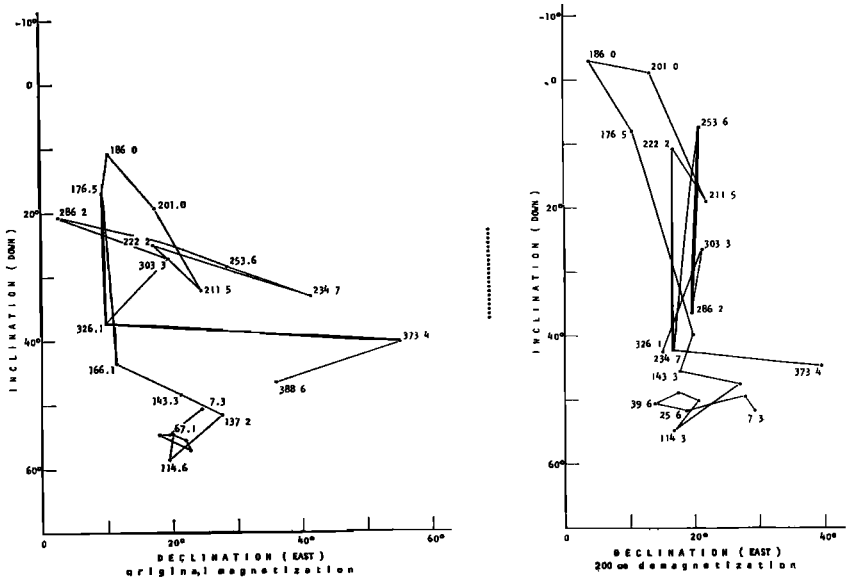
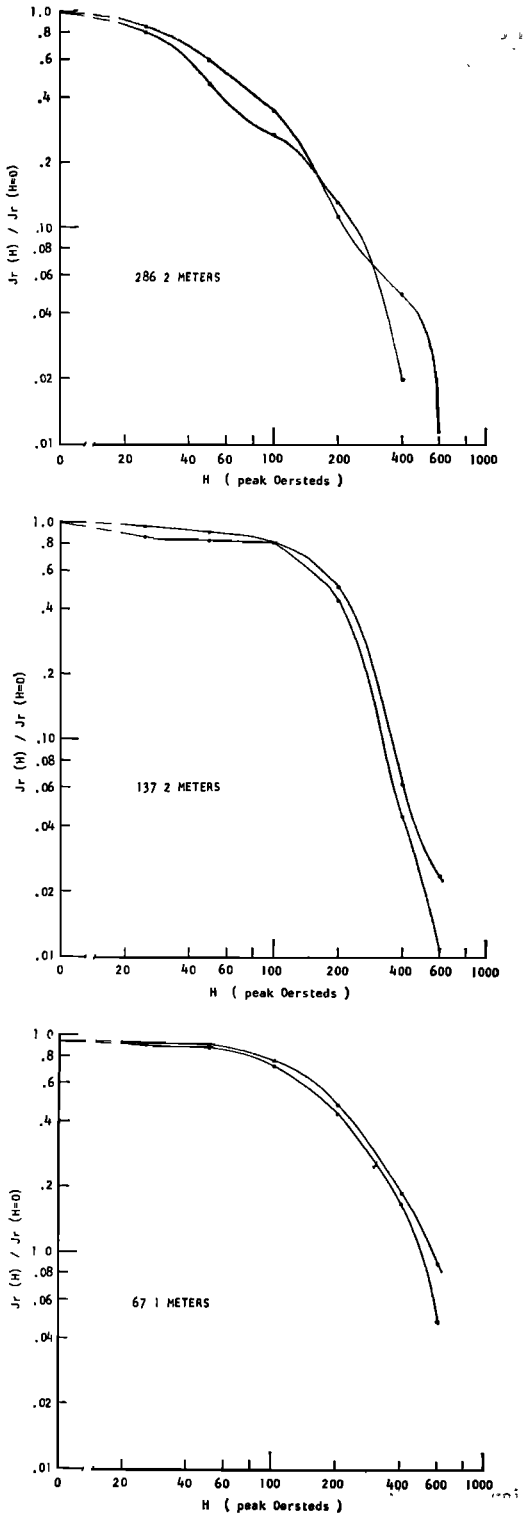


Fig. 4. Site mean direction of NRM: inclination versus declination. Site position indicated as meters above sill base.

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indicating that a comparatively large portion of the pegmatite remanence resides in the low-coercivity portion of the spectrum.

The stability of the remanent magnetization suggests an origin either as thermoremanent or as chemical remanent magnetization [*Kobayashi*, 1959]; several observations favor a thermoremanent origin. The most significant is that the pattern of magnetic directions is not related to rock type, but instead is related to position in the sill, in accord with the model for TRM acquired during cooling. Furthermore, at individual sites the magnetic directions form coherent groups, indicating the samples were all magnetized in a short time. This would be expected of TRM because only a short time would be required for the Curie temperature isotherm to pass through an individual site.

The possibility that the variation in NRM direction may be due to effects of the sill's internal magnetic field was considered. The influence of an initial high-temperature remanence on subsequent magnetization has been examined by *Strangway* [1961]. To exert a significant influence, the intensity of the initial magnetization must be in the range of 0.02 to 0.05 emu/cc, an order of magnitude greater than the maximum value at Marys Peak. The effect of the susceptibility of the body has been considered by *Strangway* [1961] and *Zidarov* [1963]. The measured susceptibilities for the Marys Peak samples range between 1 and 2 $\times$ 10^{-3} cgs units, more than an order of magnitude less than the value required to create significant field distortion.

Curie Temperatures

A necessary piece of information required to interpret the paleomagnetic results is the Curie temperature. This temperature can be expected to change through the sill. Curie temperatures were determined for whole-rock specimens from each site and polished sections cut from the same specimens were examined under reflected light by using oil immersion with magnification from 450$\times$ to 1050$\times$. The Curie temperature

Fig. 5. Demagnetization curves for representative sites. Site position indicated is distance above sill base in meters. Ordinate is the fraction of NRM remaining after treatment in alternating field of peak value H.

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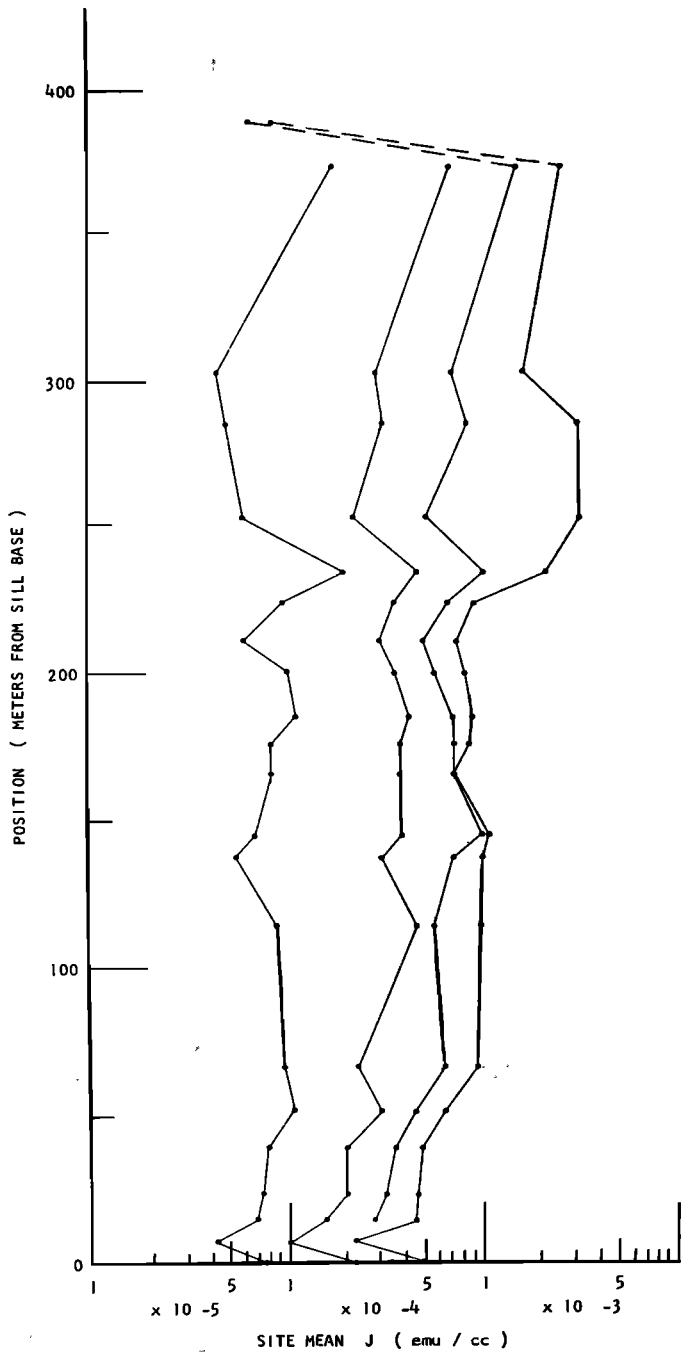


Fig. 6. Change in site mean intensity of magnetization with position in the sill and alternating-field demagnetization. Reading from right to left, values plotted are for NRM, and following demagnetization in fields of 100, 200, and 400 oe. The dashed lines connect values for the sill to those for the meta-Tyee site.

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experiments were made on an automatically recording quartz torsion balance of the Kawai type [*Clark*, 1967]. A granulated whole-rock sample was encapsuled in vacuum in a quartz vial and suspended in a noninductively wound furnace in a magnetic field of 4.2 kilogauss. Multiple tests on a Ni reference sample indicated a temperature repeatability of ±5°C and a displacement repeatability of ±10%. This accuracy is greater than that of the Curie temperature determination because of uncertainties in the slope of the magnetization curve and in the zero magnetization level. Multiple samples from the same specimen were found to have a mean standard deviation of ±9.9°C in Curie temperature.

The following paragraphs summarize the main thermomagnetic and mineralogic change through the sill.

Sill base. The rock is a coarse-grained dark granophyric gabbro, the thermomagnetization curve is flat, and the Curie temperature is

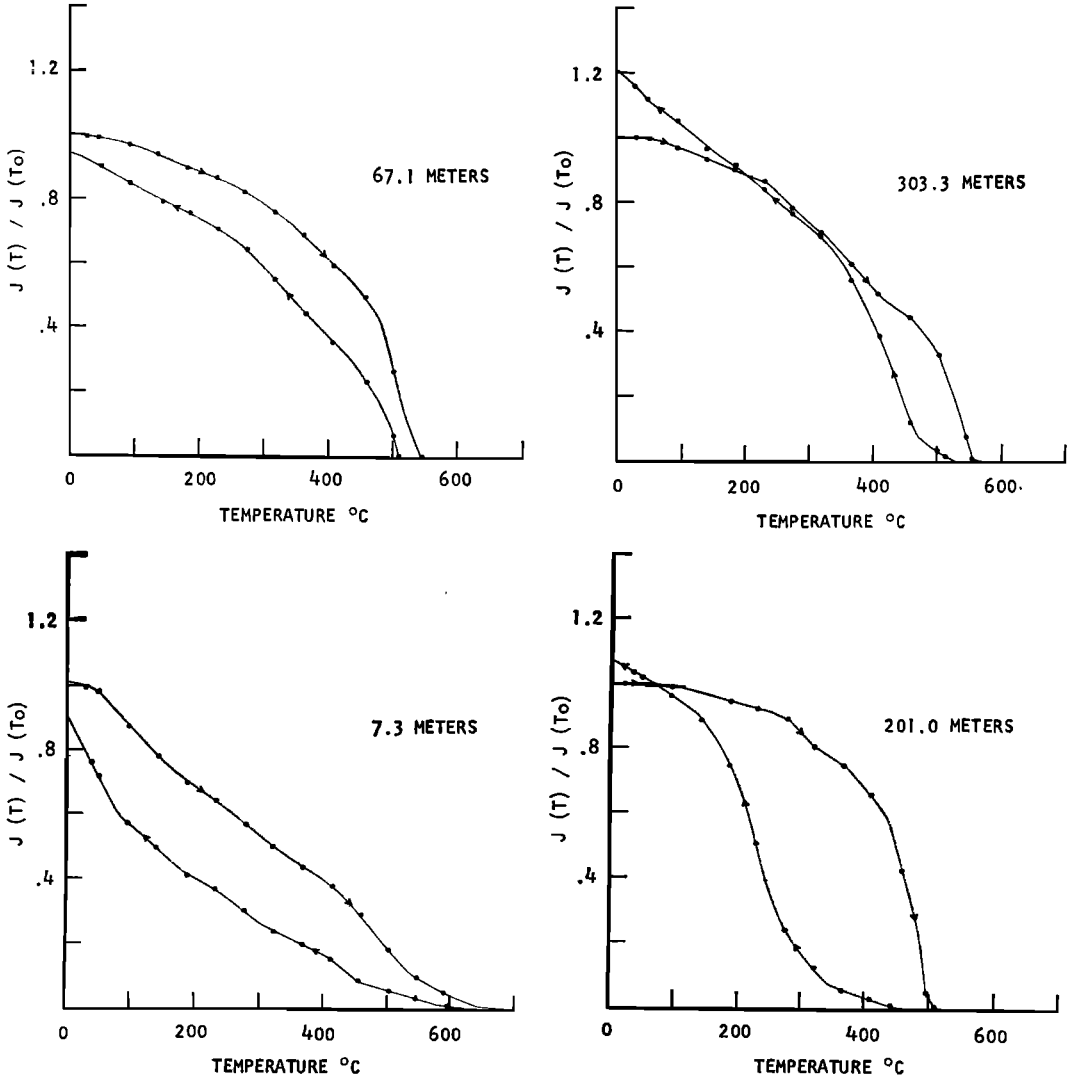


Fig. 7. Representative Curie temperature curves. Position indicated refers to elevation above sill base. The ordinate in each case is the fraction of magnetization remaining at a particular temperature relative to the magnetization at 20°C. The magnetizing field in each case was 4.2 kilogauss.

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poorly defined (Figure 7, 7.3 meters). *Akimoto* [1955] interprets this type of behavior as that of a titanomagnetite with a continuously varying composition and, therefore, a continuously varying Curie temperature. The dominant opaque mineral is a large (>300 μ) euhedral to subhedral titanomagnetite. A number of grains are composites of titanomagnetite and ilmenite.

Lower sill from 25.6 to 114.3 meters. (See Figure 7, 67.1 meters.) The heating portion of the thermomagnetization cycle traces a single, well-defined curvature whose definition improves with elevation in the gabbro facies. Curie temperatures increase from approximately 500° to 546°C. Thermomagnetic experiments made on synthetic and naturally occurring titanomagnetites [*Akimoto et al.*, 1957] indicate that an increase in Curie temperature may be due to a decrease in the titanium content of the titanomagnetite. Many of the titanomagnetite grains appear to have formed around olivine, and the opaque grain shape is increasingly controlled by the adjacent crystals.

Lower middle sill from 137.2 to 186.0 meters. The rock is a coarse-grained granophyric gabbro, lighter in color than the preceding level. The thermomagnetization curve is well defined, with Curie temperatures from 530° to 540°C, a value slightly lower than the maximum in the preceding section. A significant feature of this zone is the first appearance of ilmenite lamellas in the titanomagnetite grains, the number of lamellas increasing with elevation. Grain boundaries are broken and cracks occur through the grains.

Anomalous Curie temperature zone from 201.0 ta 211.5 meters. See Figure 7, 201.0 meters.) The rock type is a continuation of the change from granophyric gabbro to granophyric diorite and is near the gradational boundary between the two. The two sites exhibit significantly lower Curie temperatures of approximately 510°C. Ilmenite lamellas are virtually absent. The grain boundary destruction and crack development are a continuation of the trend described for the preceding section. It appears that, if conditions were such that ilmenite lamellas did not form, the titanium remained entirely in the titanomagnetite, lowering the Curie temperature to the anomalous value.

Pegmatite zone from 222.2 to approximately 320.0 meters. (See Figure 7, 303.3 meters.) This phase might be generalized as the zone of maximum Curie temperatures, approximately 550°C. According to *Akimoto's* [1957] experiments, the maximum Curie temperature of a suite may indicate a minimum fraction of titanium involved in the titanomagnetite. This interpretation for the Marys Peak measurements is supported by microscope observations, which show that the maximum concentration of ilmenite lamellas found in the sill coincide with the maximum development of the pegmatite.

Upper sill from 320.0 to 387.0 meters. This section is composed of a fine-grained granophyric diorite, locally cut by thin aplite dikes. The Curie temperatures are slightly reduced in this phase (approximately 540°C). The concentration of ilmenite lamellas in the titanomagnetite is similar to that in the pegmatite.

In summary, the Curie temperature, incidence of ilmenite lamellas, and erosion of the opaque grains increase with successive stages of differentiation. A number of investigators [*Akimoto*, 1957, 1959; *Ade-Hall*, 1964] have found a correlation between Curie temperature and oxidation. The Marys Peak sill appears to show this relationship in a single suite where oxidation associated with differentiation developed ilmenite lamellas in the titanomagnetite grains. The more highly oxidized rocks are also more strongly magnetized (Table 1 and Figure 6), as has been noted in other igneous bodies [*Balsley and Buddington*, 1958; *Watkins and Haggerty*, 1965]. However, the more highly oxidized rocks also contain a larger component of magnetization with a low coercivity, which is surprising inasmuch as strong remanence is commonly associated with high coercive force.

Paleomagnetic Record at Marys Peak

To transform the NRM measurements to a time record of the paleomagnetic field, a mathematical model approximating the sill's cooling history was developed. Limits for the amount of time involved in the record were then determined by testing the fit of the NRM data to a number of models employing various combinations of geologic parameters.

The problem is essentially the solution of the one-dimensional conductive heat flow equation with initial conditions appropriate to the Marys Peak sill after solidfication was complete. The

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cases of a slab with a uniform initial temperature and of a two-layered slab (layers of equal thickness) with a symmetric temperature distribution about its center have been treated by *Carslaw and Jaeger* [1959]. An extension of this problem applicable to the vertically differentiated Marys Peak sill is the case of an asymmetrical temperature distribution in a slab of two layers, each of arbitrary thickness. The hypothetical temperature distribution described in the geology section is summarized in Figure 8. The initial temperature distributions in the layers with thickness a and b are linear but otherwise arbitrary, and the initial temperature of the confining medium is zero. This distribution approximates that of Figure 8 (*left*) with discontinuities at the layer boundaries. The solution shows, however, that a short time after $t = 0$, the temperature distribution approaches that suggested by the geological studies.

The boundary conditions of heat flow require that the thermal flux and temperature must be continuous across each interface for all time after $t = 0$. In addition, the temperature must approach zero as x becomes very large, positive or negative.

The problem may be solved by using a Laplace transform method. The one-dimensional heat flow equation, boundary conditions, and initial conditions were transformed, and the problem was solved. The solution was then inverted term by term [*Roberts and Kaufman*, 1968]. The solution is written as

where

erfc x is defined as $(2/\pi)\int_x^\infty \exp(-\xi^2)\, d\xi$ and is known as the complementary error function.
T is temperature.
x is the space coordinate.
t is time.
k is thermal diffusivity.
r is the initial temperature of layer a at the interface.
u is the initial temperature of layer b at the interface.
q is the initial temperature gradient in layer a.
v is the initial temperature gradient in layer b.

For the Marys Peak sill, the values of a and b and the origin of the coordinate system are fixed by the dimensions of the sill and the position of the last point to solidify. Thermal diffusivity values for igneous rocks reported in the literature vary greatly [*Jaeger*, 1957; *Carslaw and Jaeger*, 1959; *Kawada*, 1964]. In addition, thermal diffusivity decreases with increasing temperature, and heating experiments on gabbros by *Kawada* [1964] suggest that the diffusivity at the temperatures of interest in this problem may be as low as half the value reported for typical measurement temperatures. Also, the effect of the heat of crystallization on the cooling history may be included as a modification of the thermal diffusivity [*Jaeger*,

For $0 < x < a$:

$$\begin{aligned}T(x, t) &= [(r + qa)/2]\ \text{erfc}\ [(a - x)/2\sqrt{kt}] \\ &+ [(u - r)/2]\ \text{erfc}\ [x/(2\sqrt{kt}] - [(u - vb)/2]\ \text{erfc}\ [(b + x)/2\sqrt{kt}] + r + qx \\ &- q\sqrt{k}/2\{2\sqrt{t/\pi}\ \exp\ [-(a - x)^2/4kt] - [(a - x)/\sqrt{k}]\ \text{erfc}\ [(a - x)/(2\sqrt{kt})]\} \\ &- [(v - q)(\sqrt{k}/2)\{2\sqrt{t/\pi}\ \exp\ [-(x)^2/4kt] - (x/\sqrt{k})\ \text{erfc}\ (x/2\sqrt{kt})\} \\ &+ (v\sqrt{k}/2)\{2\sqrt{t/\pi}\ \exp\ [-(b + x)^2/4kt] - [(b + x)/\sqrt{k}]\ \text{erfc}\ [(b + x)/2\sqrt{kt}]\} \qquad (1)\end{aligned}$$

For $0 > x > -b$:

$$\begin{aligned}T(x, t) &= [(u - r)/2]\ \text{erfc}\ (x/2\sqrt{kt}) \\ &- [(r + qa)/2]\ \text{erfc}\ [(a - x)/2\sqrt{kt}] - [(u - vb)/2]\ \text{erfc}\ [(b + x)/2\sqrt{kt}] + u + vx \\ &- q\sqrt{k}/2\{2\sqrt{t/\pi}\ \exp\ [-(a - x)^2/4kt] - [(a - x)/\sqrt{k}]\ \text{erfc}\ [(a - x)/2\sqrt{kt}]\} \\ &- (v - q)\sqrt{k}/2\{2\sqrt{t/\pi}\ \exp\ [-x^2/4kt] - x/\sqrt{k}\ \text{erfc}\ (x/2\sqrt{kt})\} \\ &+ v\sqrt{k}/2\{2\sqrt{t/\pi}\ \exp\ [-(b + x)^2/4kt] - [(b + x)/\sqrt{k}]\ \text{erfc}\ [(b + x)/2\sqrt{kt}]\} \qquad (2)\end{aligned}$$

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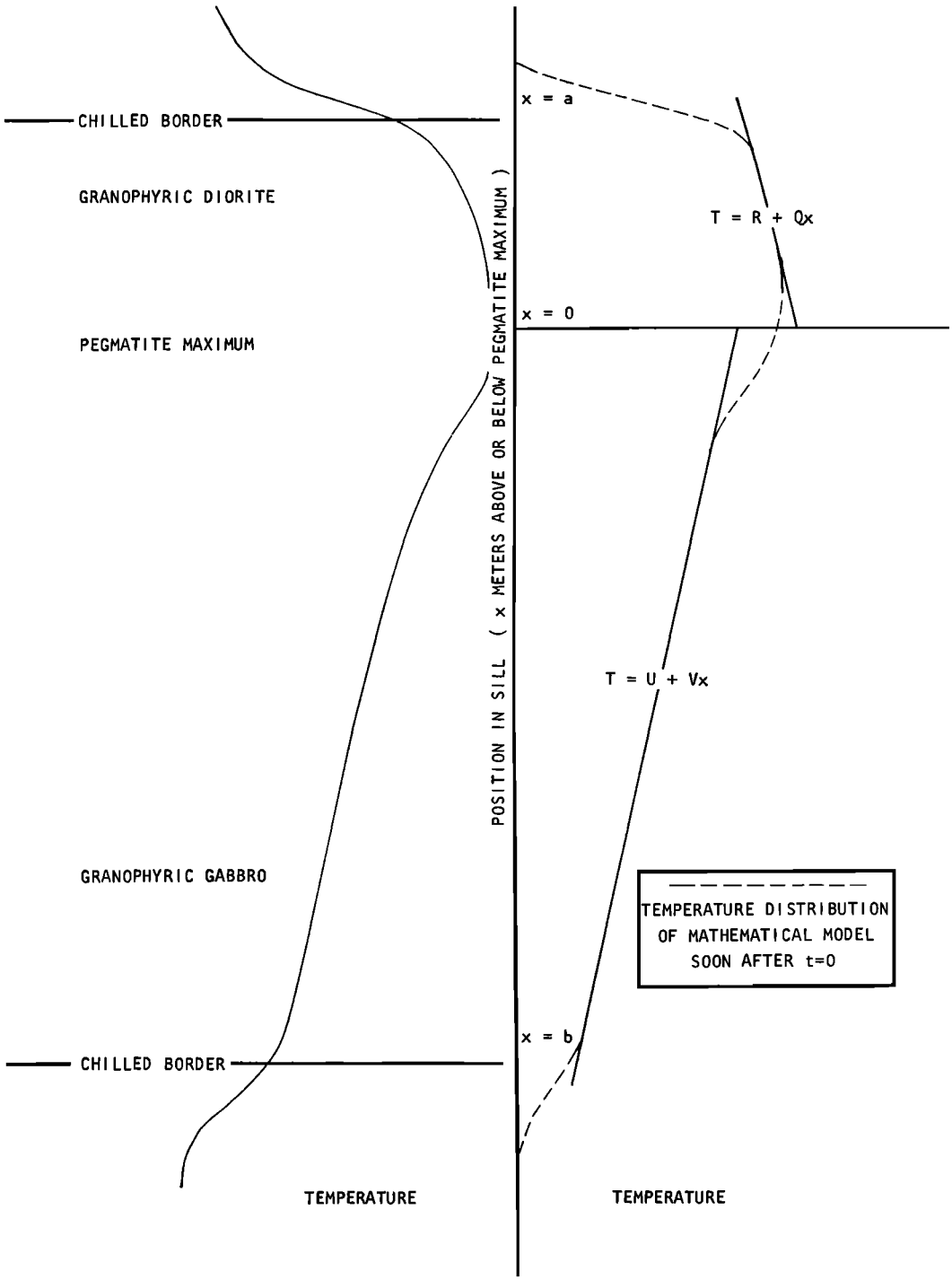
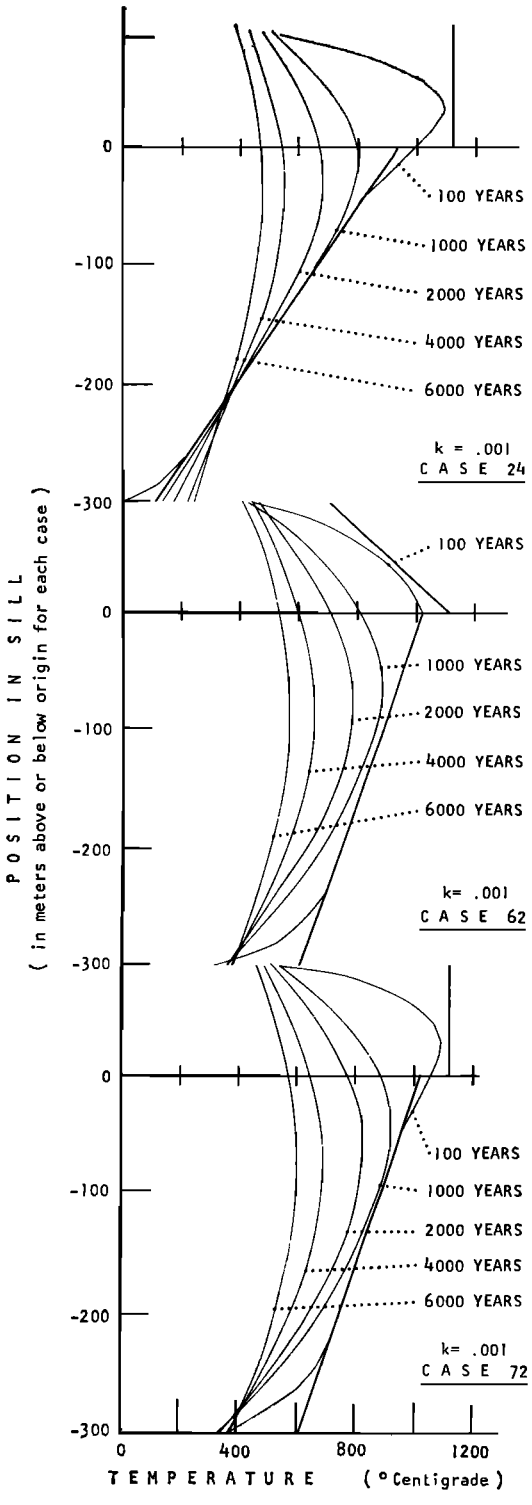


Fig. 8. Temperature distribution in a differentiated sill immediately following solidification. Abscissa is temperature increasing to right. (*Left*) Hypothetical temperature distribution according to geologic data; (*right*) approximation of the distribution for a mathematical cooling model.

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1957]. The diffusivity is related to the specific heat by

$$k = K/\rho C \qquad (3)$$

where k is thermal diffusivity, K is thermal conductivity, ρ is density, and C is the specific heat of the solid material.

The heat of crystallization is accounted for by the expression

$$C = C' + L/(\Delta T) \qquad (4)$$

where C' is the specific heat of the liquid magma, L is the latent heat of crystallization, and ΔT is the temperature range of solidification.

A range of thermal diffusivity values determined by using the published data [*Jaeger*, 1957; *Carslaw and Jaeger*, 1959; *Kawada*, 1964] and this modification is approximately 0.001 to 0.005 cm²/°C for the gabbroic material of this problem.

The NRM results provide a test for a proposed cooling history: sites with the same direction of magnetization should have acquired a magnetization at the same time. An acceptable model of the cooling history is one that permits the transformation of the NRM data to a time record that satisfies this requirement. In the lower sill the geology and similarity of NRM directions suggest that this region passed through the Curie temperature at approximately the same time. Several possible temperature configurations that might bring this about were tested, and the initial distribution that best fit the requirement was a temperature distribution in the lower sill with a low gradient near the Curie point. The NRM data also suggest that the upper sill began to acquire a magnetization later in the cooling history than the sill base did. It was found that an initial temperature distribution that concentrated high temperature in the pegmatite zone best fit this constraint. A number of cooling histories using the geometry of the Marys Peak sill and various initial temperature configurations were generated by a Burroughs B5500 digital computer. Typical histories are plotted in Figure 9. The

Fig. 9. Representative solutions of the boundary value problem for the initial temperature distributions indicated by the straight lines. The curves are temperature distributions at the indicated times after $t = 0$. k is thermal diffusivity.

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following generalities about the cooling history of a differentiated sill emerge. Regardless of the asymmetry of the original temperature distribution, the temperature tends to become symmetrically distributed about the geometrical center of the body within a few thousand years. The result of an asymmetrical initial temperature distribution is an asymmetrical migration of isotherms toward the center of the intrusion. An increase in the initial temperature difference at the interface results in an increase of the asymmetry of the isotherm migration from the top and base of the sill. The maximum asymmetry of isotherm migration for a reasonable range of differentiated sill parameters occurs with a high uniform temperature in the upper layer (a), a large temperature decrease at the interface, and a high temperature gradient in the lower layer (b).

The NRM data were transformed for several cooling history models chosen in terms of the characteristics described above. The wide range of values for the variables involved in the problem does not permit an exact solution for the cooling history, but instead suggests time limits for the paleomagnetic record. A complication in the transformation from position to time is the variation in Curie temperature through the sill. The measured Curie temperature for a site was used as the temperature of magnetization and the uncertainty of 9.9°C in Curie temperature transformed to an uncertainty in the time of magnetization. In each case, the resulting paleomagnetic record was examined to determine if sites magnetized at the same time exhibited equal NRM inclination.

Two examples of the paleomagnetic record are given here. Figure 10 is an example near the lower limit of the cooling time. The critical factor for this limit is the temperature gradient in the lower sill which will permit these sites to be magnetized at approximately the same time. This sets the lower limit of the time required for the magnetization of the sill at approximately 2000 years. Figure 11 is a paleomagnetic record typical of the upper limit of the cooling history. This limit is most sensitive to the value of thermal diffusivity. The lower limit

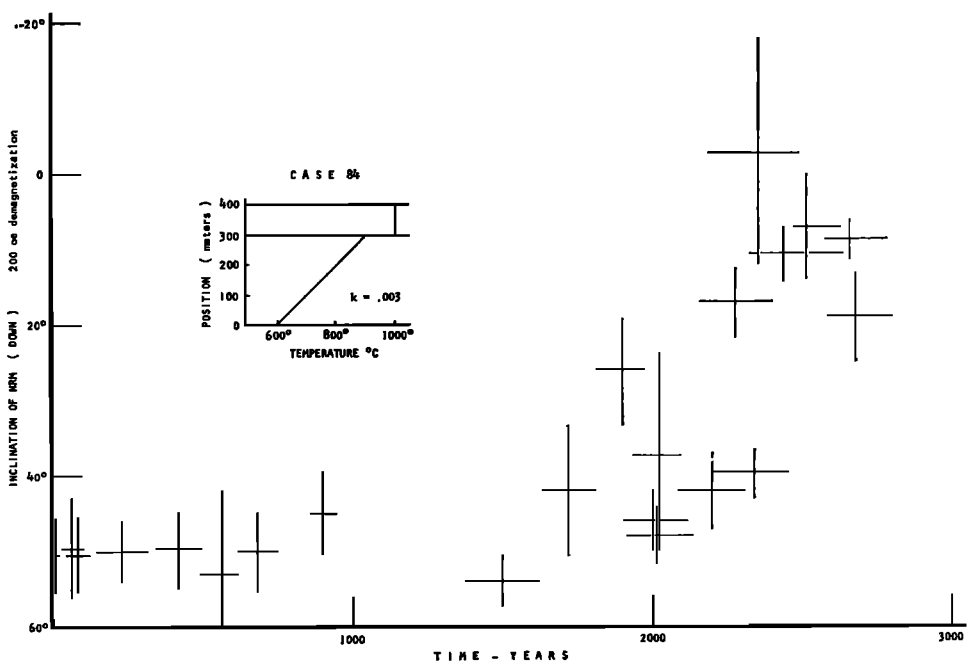


Fig. 10. Plot of time of magnetization versus inclination of NRM for a brief cooling history. Direction is site mean inclination with 95% confidence limits. The time was computed by using the Curie temperature of the site and the time error was transformed by using the mean standard deviation of the Curie temperature.

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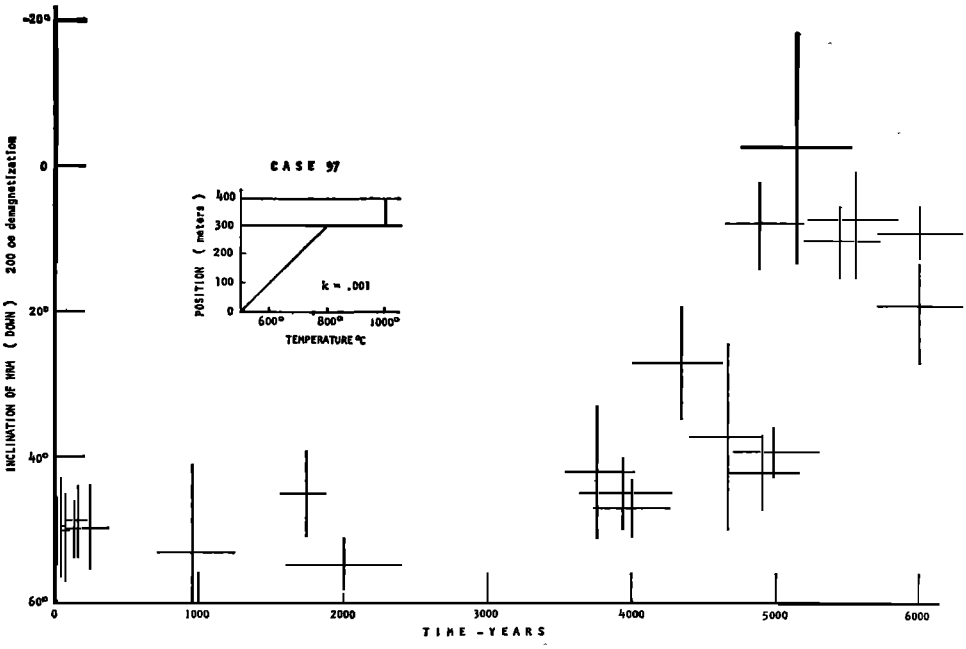


Fig. 11. Plot of time of magnetization versus inclination of NRM for a long cooling history. (See legend for Figure 10.)

of acceptable thermal diffusivity values sets the upper limit for the time of magnetization at approximately 6000 years.

Discussion

The observed change of inclination from 50° to 5° may have been produced by a change in the nondipole field. If the main dipole field had the same strength as it now has, the required intensity of the vertical component of the nondipole field required to produce this change is 0.23 gauss. The time required for the growth of this component is 2000 to 6000 years, which provides a minimum estimate of the lifetime of the process causing the growth of the nondipole field. If the observed change in direction of the field is due to a nondipole source, the source was somewhat larger than sources that exist today [*Bullard*, 1950] and almost certainly had a longer lifetime [*Cox and Doell*, 1964].

Since the change in direction is larger than would be expected in terms of the present nondipole field sources, the alteration of direction could have been due to an effect of an excursion of the main dipole field from the accepted early Tertiary position. Paleomagnetic measurements generally show the geomagnetic field to have averaged to an axial dipole during the Tertiary [*Irving*, 1964]. If the measured direction change at Marys Peak is due to a change in the orientation of the main dipole field, it represents a movement of that field of several tens of degrees from the mean Tertiary field direction over an interval of the order of thousands of years. Other investigators have reported cases of apparent Tertiary field variation of the orders of tens of degrees [*Momose*, 1963; *Watkins*, 1965; *Cox*, 1966; *Rietman*, 1966; *Heinrichs*, 1966]. The paleomagnetic record from the Marys Peak sill is of special interest because it provides an estimate of the duration of a marked departure from the field of an axial dipole.

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